

A proposal for a type trait to detect scoped enumerations

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Audience: Library Working Group

Introduction

This paper proposes `is_scoped_enum`, a new trait for the C++ Standard Library, to detect whether a type is a scoped enumeration.

Motivation and Scope

It is useful in certain contexts to know whether an enumeration is scoped or unscoped and apply (via SFINAE) different behavior depending on the type of such enumeration.

One use the author has recently worked on involves creating a set of unit tests to track the progress of a legacy library migration to modern C++. By using this trait, it is possible to define a unit test to track the progress of migration of unscoped to scoped enumerations.

Impact On The Standard

It proposes changes to an existing header, `<type_traits>`, but it does not require changes to any standard classes or functions and it does not require changes to any of the standard requirement tables.

This proposal does not depend on any other library extensions.

Naming

The existing trait to detect an enum type is `is_enum`, so variations containing

‘enumeration’ are not considered by the author.

is_scoped_enum as the suggested name was approved by LEWG in Cologne.

Wording

All proposed **additions** (there are no **deletions**) are relative to the 2020-04 working draft N4861. Editorial notes are displayed against a **gray** background

Insert into [meta.type.synop] as shown:

```
template<class T>struct is_member_pointer;

template<class T>struct is_scoped_enum;

template<class T>
inline constexpr bool is_member_pointer_v = is_member_pointer<T>::value;

template<class T>
inline constexpr bool is_scoped_enum_v = is_scoped_enum<T>::value;
```

Insert into [tab:meta.unary.prop]

template<class T>struct is_scoped_enum;	T is a scoped enumeration [dcl.enum]
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Feature test macro

Insert into [version.syn]

```
#define __cpp_lib_is_scoped_enum <DATE OF ADOPTION> // also in <type_traits>
```

Approvals

LEWG	2019-07-18
EWG	2019-07-23
CWG	2020-08-17

Example implementation

```
template<class _T, bool = is_enum_v<_T>> struct __is_scoped_enum_helper : false_type {};  
template<class _T>struct __is_scoped_enum_helper<_T, true>  
    : public bool_constant<!is_convertible_v<_T, underlying_type_t<_T>>> {};  
template<class _T>struct is_scoped_enum : public __is_scoped_enum_helper<_T> {};
```

Bibliography

[N4861] Richard Smith: “Working Draft, Standard for Programming Language C++.” ISO/IEC JTC1/SC22/WG21 document N4861(2020-04 mailing), 2020–04–01. <http://www.open-std.org/jtc1/sc22/wg21/docs/papers/2020/n4861.pdf>

Document history

Version	Date	Changes
0	2018-05-01	Initial draft
1	2020-10-12	Removed numeric sections and moved to meta.unary.prop Added feature test macro Added dates for the evolution of the paper through the working groups.